

Enterprise Google Cloud Data Platform Blueprint

VOLUME 12

Disaster Recovery and Business Continuity

Enterprise Reference Architecture and Implementation Blueprint
Architecture Baseline 1.0 | 12 July 2026

Classification: Enterprise architecture reference. Validate current service capabilities, locations, quotas, recovery features, residency constraints and contractual commitments before implementation.

Document Control

Field	Value
Document	Volume 12 - Disaster Recovery and Business Continuity
Series	Enterprise Google Cloud Data Platform Blueprint
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Status	Detailed reference architecture for adaptation and implementation
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Owner	Enterprise Business Continuity and Disaster Recovery Architecture
Approvers	CIO, CDO, CISO, Business Continuity, Enterprise Architecture, Platform Engineering, SRE, Security, Data Governance and Business Service Owners
Audience	Executives, business continuity leaders, architects, service owners, platform and data engineers, SRE, security, audit, privacy, legal and crisis communications
Review cycle	Quarterly; after major architecture or business changes; after every material incident or exercise
Primary dependencies	Volumes 01-10; Volume 11 SRE and Operations; Volume 13 Terraform and Infrastructure as Code
Scope	BIA, service tiers, RTO/RPO, failure scenarios, backup/restore, service-specific recovery, regional failover, cyber recovery, testing, continuity, communication and governance

Guidance classification. Mandatory controls apply to Tier 0, Tier 1, Restricted and Regulated workloads unless a time-bound risk exception is approved. Recommended patterns are enterprise defaults. Conditional patterns require evidence of service support, location, regulation and cost. Future-state capabilities require implementation-time validation.

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1. Executive Summary

Volume 12 defines the disaster recovery and business continuity architecture for the enterprise Google Cloud data and AI platform. It translates business impact into tiered recovery objectives and service-specific controls for a global, regulated environment operating AMER, EMEA and APAC data planes. Recovery is designed as a business-service capability, not a collection of unrelated backups.

The target state combines Google-managed service resilience with explicit customer-owned recovery controls. Tier 0 and selected Tier 1 services use paired regional data planes, replicated or protected data, pre-tested deployment artifacts, controlled standby capacity, gated automation and rehearsed incident command. Lower tiers use pilot-light or cold-rebuild patterns where the approved RTO/RPO permits. Cyber recovery is treated separately from ordinary availability failover and uses isolated clean-room recovery for high-impact destructive incidents.

No single feature is considered a complete DR solution. Replication addresses geographic loss but can replicate corruption. Backups and time travel address logical recovery but may not restore a full service. DNS redirects clients but does not restore data. Therefore every production workload must have a documented dependency graph, authoritative recovery point, write-fencing method, validation rules, decision authority, communications plan and failback procedure.

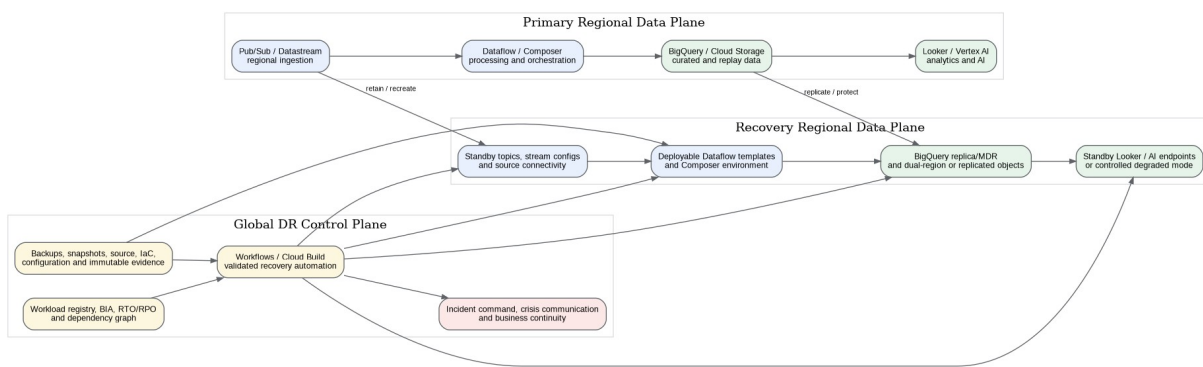


Figure 1 - Enterprise disaster recovery and business continuity reference architecture

1.1 Executive outcomes

- One hundred percent of Tier 0 and Tier 1 business services have approved BIA, owner, dependency map, RTO/RPO, recovery pattern and exercise schedule.
- Tier 0 workloads demonstrate recovery in less than one hour with less than five minutes of approved data loss, subject to documented service-specific constraints.
- Tier 1 workloads demonstrate recovery in less than four hours with less than fifteen minutes of approved data loss.
- Every protected data asset has a tested logical-recovery mechanism in addition to geographic resilience where corruption or destructive action is credible.
- Recovery privileges are segregated, time-bound and auditable; cyber recovery uses trusted restore points and clean-room controls.
- Exercises produce machine-readable evidence, findings, owners and due dates, with residual risk accepted by an accountable business authority.

2. Inherited Context, Scope and Assumptions

2.1 Inherited architecture context

- The enterprise is a global Fortune 100 organization with 10-50 PB of historical data, 50-250 TB of daily growth, billions of events and thousands of pipelines and users.
- The platform uses AMER, EMEA and APAC regional data planes, central platform services, shared governance and federated domain-owned data products.
- BigQuery is the primary analytical platform; Cloud Storage supports landing, raw, replay and archive; Pub/Sub, Dataflow, Datastream, Dataform and Managed Airflow / Cloud Composer support ingestion and engineering.
- Looker provides governed analytics; Vertex AI and Gemini support predictive and generative AI; Knowledge Catalog and Dataplex capabilities provide governance and metadata.
- Security classifications are Public, Internal, Confidential, Restricted and Regulated. VPC Service Controls, group-based IAM, workload identity, selective CMEK, centralized logging and audited administration remain inherited controls.

- The foundation defines Tier 0 RTO below one hour and RPO below five minutes; Tier 1 RTO below four hours and RPO below fifteen minutes; Tier 2 RTO and RPO below twenty-four hours.

2.2 Confirmed requirements

- Generate only Volume 12 and do not regenerate prior volumes.
- Address all 35 Volume 12 topics from the master prompt.
- Provide workload-level RTO/RPO matrices, service-specific recovery patterns, diagrams, code/configuration examples, ADRs, risks, implementation guidance and production-readiness checks.
- Use current Google Cloud terminology and mark capabilities, locations, quotas and limitations that require implementation-time validation.

2.3 Architecture assumptions

- Enterprise incident management, crisis communications, legal/privacy escalation and business continuity processes exist and can be integrated with technical recovery.
- Each production project, pipeline, dataset, dashboard and AI service has an accountable owner and is registered in a service inventory.
- Terraform and controlled CI/CD are the authoritative deployment paths; source and artifacts are retained outside the primary workload region.
- Recovery regions are selected within approved legal and data-residency boundaries and have validated service availability, connectivity, quotas and key access.
- Volume 11 monitoring/SRE capabilities provide detection, SLOs, alerting, incident command and operational evidence; this volume defines recovery architecture and exercises.

2.4 Open architecture decisions

Decision	Evidence required	Decision authority	Required before
Primary/recovery region pair per data plane	residency, latency, service availability, correlated-risk analysis, network and cost	CIO/CISO/Regional CIOs	Tier 0/1 production approval
BigQuery MDR versus dataset replication	edition eligibility, region pair, workload tests, compute capacity and commercial terms	Data Platform Architecture	Critical dataset onboarding
Cloud Storage turbo replication scope	RPO need, object write profile, supported dual-region, cost and legal approval	Storage Platform / BCDR	Tier 0 object protection
Cyber recovery isolation model	identity, key, network, evidence, backup immutability and clean-room workflow	CISO / Cyber Recovery	Regulated go-live
Looker and AI standby level	business RTO, product capabilities, content/config portability and capacity	Analytics/AI Product Councils	Tier classification
Automatic versus approval-gated failover	false-positive risk, split-brain control, business impact and reversibility	BCDR Council / Service Owner	Automation enablement

3. Disaster Recovery and Business Continuity Strategy

The strategy is to recover business services through independent, testable layers: people and decision rights; authoritative configuration and source; protected data; deployable runtime; controlled routing; validation; and communication. High availability handles localized faults. Disaster recovery handles loss of a region, dependency, control plane or trustworthy environment. Business continuity sustains critical operations while full technology service is restored.

Principle	Concrete architecture decision	Evidence
Business-first	Tier and investment derive from BIA and business-owner approval	BIA, tier, financial/regulatory impact
Design for failure independence	Recovery plane uses a separate region and, for cyber recovery, separate projects/identities/keys	failure-domain analysis and clean-room design
Protect against logical and physical loss	Use replication plus time travel/snapshots/soft	restore tests and retention policy

Principle	Concrete architecture decision	Evidence
	delete/versioning and source-based rebuild	
Automate deterministic steps	Use IaC and workflows with approvals, fencing and validation	pipeline logs, approvals and evidence bundle
Fail closed on integrity	Do not release traffic until reconciliation, security and dependency checks pass	validation report and owner sign-off
Exercise the full sociotechnical system	Test people, decisions, suppliers, communications and technology	exercise report and remediation register
Respect sovereignty	Recovery copies and processing remain within legally approved locations	residency matrix and legal approval
Make degraded service explicit	Document read-only, delayed, cached, alternate-channel or manual workarounds	business continuity plan

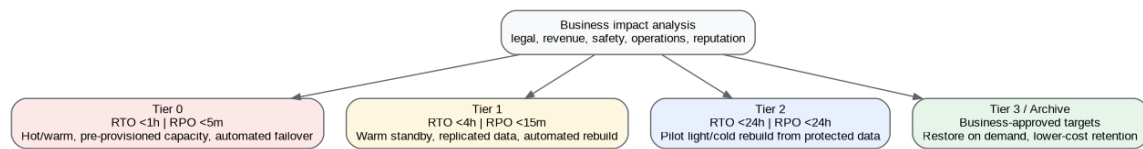


Figure 2 - Business-impact-driven recovery patterns by service tier

4. Business Impact Analysis

The BIA evaluates loss of confidentiality, integrity and availability together with financial, customer, safety, legal, regulatory, operational and reputational impact. The output is a business-service record, not merely a technical application record. One business service can depend on many data products, pipelines, identities, keys, networks, SaaS systems and human processes.

BIA dimension	Questions	Tier escalation indicator
Revenue/customer	What revenue, orders, service commitments or customer journeys stop?	material customer harm or contractual breach within hours
Regulatory/legal	Does outage or data loss breach reporting, records, privacy or market obligations?	mandatory deadline or regulated evidence at risk
Safety/clinical/operational	Could delayed or incorrect data affect safety, care, operations or supply continuity?	human or operational safety impact
Data integrity	Can incorrect or duplicated data drive irreversible decisions?	financial posting, treatment, inventory or compliance decisions
Dependency centrality	How many downstream products and regions depend on the service?	shared platform or authoritative enterprise source
Workaround	Is a manual, cached, read-only or alternate-channel process viable?	no viable workaround within impact tolerance
Recovery complexity	Are source logs, external partners, keys or specialist skills required?	long lead-time dependency or scarce expertise

4.1 BIA acceptance criteria

- Business owner, service owner, data owner, SRE, security and continuity representatives approve the record.
- Maximum tolerable period of disruption and maximum tolerable data loss are expressed in business terms before technical targets are assigned.
- Critical dependencies include identity, KMS, networking, DNS, source systems, external SaaS/partners, metadata, CI/CD and human roles.
- Degraded-mode procedures identify capacity, staffing, data limitations, customer impact and exit criteria.
- BIA is reviewed at least annually and after material business, regulatory, architecture or dependency changes.

5. Service-Tier Classification, RTO and RPO

RTO starts when the declared service disruption begins and ends when the business service is restored to the agreed minimum operating level. RPO measures the maximum acceptable loss of committed business data, not only

infrastructure replication lag. Recovery time must include detection, decision, fencing, restore, validation, traffic release and business confirmation.

Tier	Inherited target	Default pattern	Minimum exercise
Tier 0	RTO <1 hour; RPO <5 minutes	Hot or warm paired-region recovery; protected data and pre-provisioned critical capacity; gated automation	Annual end-to-end regional exercise plus quarterly component restores
Tier 1	RTO <4 hours; RPO <15 minutes	Warm standby or automated rebuild; replicated/protected data; capacity and dependency validation	Annual or risk-based regional exercise; semiannual component failover
Tier 2	RTO <24 hours; RPO <24 hours	Pilot light or cold rebuild from IaC, source and protected data	Annual restore and runbook test
Tier 3	Business-approved archive targets	Restore on demand from governed retention copies	Periodic sample restore based on regulation and risk

5.1 Workload-level RTO/RPO matrix

ID	Business service	Tier	RTO	RPO	Recovery pattern
DR-PLAT-001	Enterprise data access and curated Tier 0 products	Tier 0	<1 hour	<5 minutes	Warm/hot recovery plane; BigQuery MDR or proven replica pattern; pre-provisioned critical capacity
DR-ING-001	Regulated transaction CDC and event ingestion	Tier 0	<1 hour	<5 minutes	Source HA plus retained/replayable events; pre-created secondary connectivity and templates
DR-ANL-001	Executive and operational analytics	Tier 1	<4 hours	<15 minutes	Replicated datasets, deployable Looker configuration, warm capacity
DR-AI-001	Customer-facing AI inference grounded on enterprise data	Tier 1	<4 hours	<15 minutes for governed state	Region-paired deployment config, protected model artifacts, degraded-mode routing
DR-ORCH-001	Cloud Composer orchestration for critical pipelines	Tier 1	<4 hours	<15 minutes metadata target where feasible	Scheduled snapshots in cross-region-redundant bucket; pre-tested environment rebuild
DR-GOV-001	Metadata, glossary, lineage and access-policy evidence	Tier 1	<4 hours	<15 minutes for policy/configuration changes	Configuration as code, exports where supported, authoritative source registry
DR-BI-001	Standard BI and departmental reporting	Tier 2	<24 hours	<24 hours	Cold/warm rebuild from source control and replicated curated data
DR-DEV-001	Non-production engineering environments	Tier 2	<24 hours	<24 hours	Recreate from IaC and source; no dedicated hot standby
DR-ARC-001	Historical archive and legal records	Tier 3	Business-approved, normally 72 hours	Policy-defined	Protected multi/dual-region objects, retention lock where required, restore on demand

The complete owner-inclusive matrix is provided in `matrices/workload_rto_rpo_matrix.csv`. Workloads cannot inherit a stricter tier merely because a dependent platform has one; the business owner must approve the service objective and fund the required pattern.

6. Failure Scenarios

Architecture reviews must cover at least five failure families: regional, zonal, dependency, data corruption and cyber. The response differs by failure type. Invoking a regional failover for a logical corruption can replicate or expose the same bad state in the recovery region. Conversely, restoring an old backup during a transient service outage can unnecessarily increase data loss.

Scenario	Detection and decision	Recovery response	Key control
Regional service outage	Correlated provider/service errors plus business impact; disaster declaration by incident commander	Fence primary, validate recovery state, activate paired region and redirect consumers	avoid split brain and confirm capacity
Zone failure	Managed-service and infrastructure health signals	Use multi-zone service behavior or reschedule; escalate only if blast radius expands	distinguish HA from DR
Dependency failure	Identity, KMS, DNS, network, source, partner or SaaS SLO breach	Queue/retry, alternate approved path, degraded mode or dependency-specific failover	dependency graph and ownership
Data corruption	Quality, reconciliation, lineage or user report	Stop propagation, preserve evidence, restore clean point, reprocess and reconcile	authoritative recovery point
Cyber incident	SCC/SIEM, audit, integrity or threat detection	Isolate, revoke, preserve evidence, clean-room restore and security approval	trusted recovery environment



Figure 3 - Clean-room cyber recovery sequence

7. Backup Strategy

The backup strategy is policy-based and workload-specific. Managed-service durability is not treated as a backup. Replication is not treated as an independent recovery point. Critical services use multiple protection layers selected from time travel, snapshots, clones, soft delete, versioning, retention policy, cross-region replication, model/artifact retention, environment snapshots, source control and immutable release artifacts.

Control objective	Mandatory design rule
Recover accidental deletion/overwrite	Use service logical recovery features and test restore within their retention window.
Recover corruption	Maintain recovery points that predate propagation and record lineage/checkpoints.
Recover regional loss	Use supported cross-region or dual-region patterns within approved residency.
Recover cyber destruction	Use isolated, access-controlled, tamper-resistant copies or exports and separate recovery identities.
Recover configuration	Store declarative configuration, source and immutable release artifacts outside runtime state.
Meet legal/records obligations	Apply retention lock/hold only through records approval; preserve privacy deletion and minimization requirements.
Prove recoverability	Run sample restores, hash/control-total validation and evidence capture on a defined cadence.

7.1 Backup catalog

Asset	Mechanism	Frequency	Recovery location	Owner
BigQuery curated Tier 0	Replication/MDR plus periodic snapshots for selected high-value tables	Continuous + daily snapshots	Paired allowed region	Data product owner
Cloud Storage raw/replay	Dual-region or replicated	Continuous + policy based	Residency-approved	Storage platform

Asset	Mechanism	Frequency	Recovery location	Owner
	bucket; soft delete; optional versioning; retention/legal hold		geography	
Pub/Sub replay	Topic/subscription retention and pre-change snapshot	Continuous + before high-risk changes	Regional service	Streaming platform
Composer	Scheduled environment snapshot	At least every 15 minutes for Tier 1 or evidence-based cadence	Cross-region-redundant bucket	Orchestration platform
Looker	Managed backup plus LookML in Git; content export where required	Daily + before change	Per product support and approved recovery location	Analytics platform
Vertex AI	Model artifacts, containers, pipeline source, prompt/config versions and evaluation evidence	Every release	Artifact Registry/Cloud Storage/Git in approved region	AI platform
Metadata and glossary	Definitions/configuration as code and periodic inventory/export where supported	Every change + daily inventory	Central protected repository	Governance platform
Terraform state	State versions, soft delete/versioning and controlled export	Every apply	Protected state bucket + approved recovery copy	Cloud platform
Source and CI/CD	Git provider redundancy/export, immutable artifacts and build provenance	Every commit/release	Enterprise source and artifact platforms	DevSecOps

8. Restore Strategy and Recovery Sequencing

Recovery follows dependency order. Identity, keys, network and policy must be available before data services. Data and metadata must be trustworthy before pipelines are started. Pipelines must be fenced and reconciled before consumers are redirected. Business owners release the service only after minimum operating capability and integrity checks pass.

Sequence	Action	Exit criterion
0	Declare incident, assign commander, preserve evidence and open change record	decision authority and communications active
1	Fence primary writes, schedulers, consumers and automated retries where necessary	no uncontrolled dual processing
2	Validate identity, keys, network, service perimeter, quotas and recovery-region health	foundational dependencies pass
3	Select trusted recovery point and establish expected RPO/data gap	recovery point approved
4	Restore or activate data and metadata	integrity and access checks pass
5	Start ingestion and processing in controlled order	backlog/watermark progressing without unacceptable duplicates
6	Restore analytics, APIs and AI endpoints or activate degraded mode	technical SLO and security checks pass
7	Reconcile business totals, approve traffic release and communicate status	business owner accepts minimum service
8	Maintain heightened monitoring, evidence and decision log	stable operation and next checkpoint agreed



Figure 4 - Controlled regional failover sequence

9. BigQuery Recovery

BigQuery recovery uses different mechanisms for logical errors and hard regional failure. Time travel and fail-safe retain changed or deleted data for recovery windows. Table snapshots and clones provide explicit points for high-risk changes and regulated evidence. Cross-region dataset replication or BigQuery Managed Disaster Recovery can support regional outage recovery. Managed DR requires implementation-time validation of eligible editions, locations, features, quotas, failover behavior and commercial terms.

Failure	Primary pattern	Validation
Accidental table update/delete	Time travel, table copy from point in time, snapshot or clone	row counts, checksums/control totals, schema, policy tags, row/column security
Bad pipeline release	Stop writes, restore affected tables or rebuild from immutable source/replay	lineage checkpoint, duplicate and financial control totals
Dataset deletion	Use supported undelete/recovery capability within window or recreate from protected copies	dataset IAM, metadata and downstream object references
Regional outage	Managed DR or cross-region dataset replica; activate compute and consumer configuration	replication lag/RPO, reservations/capacity, authorized views, routines and consumer endpoints
Cyber incident	Recover clean point into isolated project/dataset and reapply policy from source	malware/configuration assurance, least privilege and business reconciliation

9.1 BigQuery implementation controls

- Classify datasets by tier and decide which tables require explicit snapshots in addition to replication.
- Replicate or recreate dependent routines, views, authorized views, policies, connections and reservation assignments; data copy alone is insufficient.
- Use separate recovery identities and a consumer cutover gate. Do not publish a replica merely because it is queryable.
- Measure actual RPO from the last committed business record available in recovery, not from the time the failover command completed.
- Validate cross-region data transfer, storage and standby compute cost through Volume 10 FinOps governance.

9.2 Representative recovery script

```
#!/usr/bin/env bash
# Purpose: controlled BigQuery recovery starter using table snapshots or a replicated dataset.
# Prerequisites: bq CLI, authenticated recovery identity, approved incident and tested target datasets.
set -euo pipefail

: "${INCIDENT_ID:?INCIDENT_ID required}"
: "${PROJECT_ID:?PROJECT_ID required}"
: "${PRIMARY_DATASET:?PRIMARY_DATASET required}"
: "${RECOVERY_DATASET:?RECOVERY_DATASET required}"
: "${LOCATION:?LOCATION required}"
: "${RECOVERY_TIMESTAMP:?RECOVERY_TIMESTAMP required, e.g. 2026-07-12T04:00:00Z}"

log() { printf '%s incident=%s %s\n' "$(date -u +%FT%TZ)" "$INCIDENT_ID" "$*"; }

log "Validating datasets"
bq --location="$LOCATION" show --dataset "${PROJECT_ID}:${PRIMARY_DATASET}" >/dev/null
bq --location="$LOCATION" show --dataset "${PROJECT_ID}:${RECOVERY_DATASET}" >/dev/null

# Example critical table list is read from a governed file, not hard-coded in production.
while IFS=, read -r table_name expected_min_rows; do
  [[ "$table_name" == "table_name" ]] && continue
  log "Creating recovery snapshot for ${table_name}"
```

10. Cloud Storage Recovery

Cloud Storage protection combines geographic placement, logical deletion recovery and records controls. Dual-region or multi-region storage provides geographic redundancy. Turbo replication can be selected for eligible dual-region buckets requiring a fifteen-minute replication RPO. Soft delete retains deleted or overwritten objects for a configured period. Object Versioning can preserve generations where workload behavior and cost justify it. Retention policies and holds protect records but require careful governance because they can prevent deletion, including privacy deletion.

Data class/use	Default pattern	Additional controls
Tier 0/1 raw and replay	Residency-approved dual-region or explicit paired replication	turbo replication where approved; soft delete; versioning for critical mutable objects; CMEK
Curated exports and model artifacts	Cross-region-redundant bucket or controlled replication	integrity hashes, release manifest and immutable naming
Regulated records	Approved location plus retention policy/lock and legal holds	records owner approval, deletion exception and access evidence
Tier 2 working data	Regional bucket plus source-based rebuild or scheduled copy	soft delete and lifecycle policy
Cyber recovery evidence	Separate restricted project/bucket and recovery identity	retention, tamper resistance, access monitoring and chain of custody

10.1 Representative Terraform

```
# Purpose: protected dual-region recovery bucket for critical objects.
# Validate region pair, residency, feature support, quotas and organization policies before use.
variable "project_id" { type = string }
variable "bucket_name" { type = string }
variable "dual_region" { type = string }
variable "kms_key_name" { type = string }

resource "google_storage_bucket" "dr" {
  project      = var.project_id
  name         = var.bucket_name
  location     = var.dual_region
  storage_class = "STANDARD"
  uniform_bucket_level_access = true
  public_access_prevention = "enforced"
  force_destroy = false

  rpo = "ASYNC_TURBO"

  versioning { enabled = true }

  soft_delete_policy {
    retention_duration_seconds = 1209600 # 14 days; replace with approved class policy.
  }

  retention_policy {
    retention_period = 2592000 # 30 days; do not lock until governance approval.
  }
}
```

11. Pub/Sub Replay and Streaming Recovery

Pub/Sub reliability decouples producers and consumers, but recovery must be explicitly designed. Topic or subscription retention preserves messages for a configured window. Snapshots capture subscription acknowledgment state and seek repositions a subscription to a snapshot or timestamp. Replay can produce duplicate or out-of-order business effects, so sinks and transformations must implement idempotency, deterministic business keys, checkpointing and reconciliation.

Control	Implementation requirement
Retention	Set topic/subscription retention long enough for the Tier RTO plus detection, decision and replay duration.
Snapshot	Create before high-risk releases and maintain an automated cadence for selected Tier 0/1 subscriptions.
Fencing	Pause or isolate current consumers before seek or recovery-subscription activation.
Replay	Prefer a new recovery subscription for controlled reprocessing where design permits.
Idempotency	Use stable event IDs/business keys, deduplication tables or idempotent merge semantics.

Control	Implementation requirement
Validation	Measure oldest unprocessed event, backlog, duplicate rate, dead letters, freshness and business control totals.

11.1 Representative replay script

```
#!/usr/bin/env bash
# Purpose: snapshot and replay a Pub/Sub subscription for controlled pipeline recovery.
set -euo pipefail
: "${PROJECT_ID:?}"; : "${SUBSCRIPTION:?}"; : "${SNAPSHOT:?}"; : "${INCIDENT_ID:?}"

log(){ printf '%s incident=%s %s\n' "$(date -u +%FT%TZ)" "$INCIDENT_ID" "$*"; }

gcloud pubsub subscriptions describe "$SUBSCRIPTION" --project "$PROJECT_ID" >/dev/null
if ! gcloud pubsub snapshots describe "$SNAPSHOT" --project "$PROJECT_ID" >/dev/null 2>&1; then
  log "Creating snapshot ${SNAPSHOT}"
  gcloud pubsub snapshots create "$SNAPSHOT" --subscription "$SUBSCRIPTION" --project "$PROJECT_ID"
fi

log "Seeking subscription to snapshot. Confirm consumers are fenced before this step."
gcloud pubsub subscriptions seek "$SUBSCRIPTION" --snapshot "$SNAPSHOT" --project "$PROJECT_ID"
log "Seek submitted. Monitor backlog, duplicates, processing errors and business reconciliation."
```

12. Dataflow Recovery

Dataflow jobs are regional. Recovery is performed by starting an approved pipeline artifact in the recovery region and coordinating source state, replay position, sink state and write fencing. Flex Templates, container images, dependencies, pipeline options, KMS keys, service accounts, subnet configuration and monitoring must be reproducible from source and IaC. Stateful streaming recovery requires workload-specific testing; do not assume runtime state transfers across regions.

- For Pub/Sub sources, use the replay workflow and a controlled subscription. For file sources, use immutable object manifests and processed-file checkpoints.
- For BigQuery sinks, use idempotent MERGE or business-key deduplication and isolate recovery writes until reconciliation passes.
- Drain versus cancel is an incident decision. Draining preserves in-flight work but can exceed RTO; cancellation can require replay.
- If Streaming Engine and CMEK are used, validate key availability and snapshot/recovery behavior before disabling or rotating keys.
- Success criteria include job health, watermark advance, source backlog reduction, duplicate rate, schema compatibility, output reconciliation and cost/capacity stability.

12.1 Representative Dataflow recovery starter

```
#!/usr/bin/env bash
# Purpose: start a tested Dataflow Flex Template in the recovery region.
set -euo pipefail
: "${PROJECT_ID:?}"; : "${REGION:?}"; : "${TEMPLATE_SPEC_GCS_PATH:?}"; : "${JOB_NAME:?}"
: "${SERVICE_ACCOUNT_EMAIL:?}"; : "${SUBNETWORK:?}"; : "${TEMP_LOCATION:?}"; : "${STAGING_LOCATION:?}"
: "${INPUT_SUBSCRIPTION:?}"; : "${OUTPUT_TABLE:?}"; : "${INCIDENT_ID:?}"

params="inputSubscription=${INPUT_SUBSCRIPTION},outputTable=${OUTPUT_TABLE},incidentId=${INCIDENT_ID}"

gcloud dataflow flex-template run "$JOB_NAME" \
  --project "$PROJECT_ID" \
  --region "$REGION" \
  --template-file-gcs-location "$TEMPLATE_SPEC_GCS_PATH" \
  --service-account-email "$SERVICE_ACCOUNT_EMAIL" \
  --subnetwork "$SUBNETWORK" \
  --temp-location "$TEMP_LOCATION" \
  --staging-location "$STAGING_LOCATION" \
  --parameters "$params" \
  --disable-public-ips

# Deployment success is not recovery success. Validate backlog, duplicate rate,
# watermark, output reconciliation, KMS access and SLO before traffic release.
```

13. Datastream Recovery

Datastream streams are regional. During a regional outage, a stream can remain unavailable until the region recovers and then continue from its last position; duplicate destination events can occur. Stream recovery can support source database failover, unavailable log positions or prolonged connectivity loss. The design must preserve source logs for the required RPO, maintain secondary connectivity and profiles, and reconcile the destination after recovery.

Scenario	Action	Data control
Datastream regional outage	Wait for recovery or execute approved paired-region recreation if feasible	track last source position and destination timestamp
Source database failover	Update/recover source profile using stream recovery procedure	confirm new source continuity and backfill gap
Source logs aged out	Recover from first available position and backfill missing interval from source snapshot/export	deduplicate and reconcile destination
Destination unavailable	Pause/retry and preserve source log window; restore destination first	avoid losing source changes during extended outage
Credential/key issue	Restore secret/KMS/IAM dependency through security-approved process	audit access and rotate compromised credentials

14. Managed Airflow / Cloud Composer Recovery

Environment snapshots protect orchestration metadata and configuration and can be loaded to restore an environment or create a new environment at the same or later supported version. Scheduled snapshots should be written to a separate bucket that remains available during a regional outage. Snapshot contents include the Airflow database and fernet material, so the bucket is a high-sensitivity asset. Secrets used by DAGs should be externalized to Secret Manager rather than embedded in Airflow connections.

- Use scheduled snapshots at a cadence aligned to metadata RPO and create manual snapshots before high-risk upgrades.
- Pause DAGs before snapshot-based migration or failover to avoid duplicate scheduled runs.
- Keep DAG source, requirements, plugins, variables policy and environment configuration in Git/IaC; snapshots are not the only source of truth.
- Pre-create network, service account, KMS, bucket, secret and private connectivity dependencies in the recovery region or automate them through tested modules.
- After restore, validate DAG inventory, schedules, connections, variables, pools, task state, duplicate-run controls and downstream access.

14.1 Representative snapshot script

```
#!/usr/bin/env bash
# Purpose: save or load a Managed Airflow / Cloud Composer environment snapshot.
set -euo pipefail
ACTION=${1:?save|load}
: "${PROJECT_ID:?}"; : "${ENVIRONMENT:?}"; : "${LOCATION:?}"; : "${SNAPSHOT_LOCATION:?}"

case "$ACTION" in
  save)
    gcloud composer environments snapshots save "$ENVIRONMENT" \
      --project "$PROJECT_ID" --location "$LOCATION" \
      --snapshot-location "$SNAPSHOT_LOCATION"
    ;;
  load)
    : "${SNAPSHOT_PATH:?exact snapshot path required for load}"
    gcloud composer environments snapshots load "$ENVIRONMENT" \
      --project "$PROJECT_ID" --location "$LOCATION" \
      --snapshot-path "$SNAPSHOT_PATH"
    ;;
  *) echo "Unsupported action" >&2; exit 64;;
esac
# Pause DAGs and control scheduling during migration/recovery to avoid duplicate runs.
# Snapshot locations contain sensitive Airflow database content and fernet material.
```

15. Looker Recovery

Looker recovery separates application instance recovery, LookML source recovery and user-created content recovery. Looker (Google Cloud core) backups can restore to the same instance; moving operational content to another instance requires supported import/export procedures. LookML must remain in enterprise Git and deploy through controlled branches. A Tier 1 design therefore includes instance configuration as code where supported, Git-backed LookML, scheduled backups, content export strategy, dependency mapping and a tested recovery instance pattern.

Asset	Protection	Recovery test
LookML projects	Enterprise Git, protected branches, signed releases	deploy known release to recovery instance
Instance operational data	Managed automatic/manual backups	same-instance restore test within supported process
Cross-instance content	Supported import/export and content inventory	import to recovery instance and validate folders, schedules and permissions
Connections and secrets	Secret Manager/configuration records; least privilege	validate database connectivity and service accounts
Embeds and routing	Application configuration and DNS/load-balancer controls	test authentication, signed embed and consumer cutover
Elite System Activity or non-backed telemetry	Independent export/retention if required	confirm monitoring continuity and known backup exclusions

16. Vertex AI and Gemini Recovery

AI recovery must protect more than a deployed endpoint. The recoverable product includes training data references, code, containers, model artifacts or registered versions, feature definitions, prompts, retrieval configuration, evaluation datasets, safety settings, IAM, KMS, networking, quotas and application routing. Supported locations and endpoint behavior vary by model and capability, so the paired-region design is validated per AI product.

AI component	Recovery pattern	Validation
Custom model artifacts	Store versioned artifacts/container digests in approved redundant storage and registry	checksum, provenance and framework/runtime compatibility
Model registry metadata	Maintain release manifest with model version, artifact URI, evaluation and deployment parameters	recreate or import in recovery region
Online endpoint	Pre-create or deploy endpoint in paired region based on tier; route through application control plane	quota, latency, autoscaling, private access and prediction parity
Generative AI configuration	Version prompts, model aliases, safety settings, tools and evaluation data	quality, safety, residency and fallback model tests
RAG index/data store	Protect source corpus and ingestion manifests; rebuild or maintain supported regional copy	document count, permissions, retrieval quality and citations
Agent/tool integration	Version agent definition and tool contracts; maintain degraded mode	identity, authorization, side-effect control and human oversight

For business-critical inference, the application should support a tested secondary deployment or approved degraded mode. Do not assume that a global endpoint alone satisfies residency, deterministic failover or application-state recovery. Validate current product documentation, supported regions, quotas and data handling before production approval.

17. Metadata Recovery

Metadata is a control-plane dependency. Recovery must restore business definitions, ownership, classification, policy tags, data-product records, contracts, quality rules and lineage configuration sufficiently to prevent uncontrolled access or ambiguous data use. Managed catalog availability does not remove the need for authoritative configuration and export/inventory evidence.

- Maintain glossary terms, domain/product definitions, data contracts, quality rules and policy configuration in version-controlled repositories or governed APIs where supported.

- Export periodic asset inventories, tag templates, policy references and lineage/source mappings where supported and useful.
- Rebuild discovery scans, profiles and quality jobs through IaC or deployment scripts rather than manual recreation.
- After recovery, validate ownership, classification, policy tags, masking/row-level controls, lineage completeness and certification status before broad consumer access.
- If metadata is unavailable, enforce a restrictive degraded mode: existing approved access may continue only where risk accepted; new access and certification are suspended.

18. Terraform State, Source Code and Configuration Recovery

Infrastructure and configuration recovery is a Tier 0 platform capability because every service rebuild depends on it. State, source and release artifacts must not share the same failure domain or administrator path as the workload being recovered. Recovery procedures preserve state serial integrity and reconcile restored state with actual resources before any apply operation.

Asset	Protection	Recovery control
Terraform state	Dedicated protected Cloud Storage bucket, versioning/soft delete, CMEK, restricted IAM and approved recovery copy/export	restore exact generation; inspect serial; refresh-only plan; peer approval before apply
Source code	Enterprise Git with protected branches, replicated/exported repository and off-region access	validate commit/tag signatures and dependencies
Build artifacts	Immutable Artifact Registry images/packages and release manifests	verify digest, provenance and vulnerability status
Service configuration	Terraform, YAML/JSON, SQL, LookML, DAGs, templates and policy as code	deploy known-good tagged release
Secrets and keys	Secret Manager and Cloud KMS recovery architecture; documented break-glass and key dependencies	security authorization, key availability and rotation after compromise
Certificates/DNS	Certificate and DNS configuration under controlled source/IaC	validate trust chain, TTL and routing after change

18.1 Representative state-bucket Terraform

```
# Purpose: hardened Terraform state bucket. Apply in a dedicated bootstrap project.
variable "project_id" { type = string }
variable "bucket_name" { type = string }
variable "region" { type = string }
variable "kms_key_name" { type = string }

resource "google_storage_bucket" "tfstate" {
  project           = var.project_id
  name              = var.bucket_name
  location          = var.region
  storage_class     = "STANDARD"
  uniform_bucket_level_access = true
  public_access_prevention = "enforced"
  force_destroy     = false
  versioning { enabled = true }
  soft_delete_policy { retention_duration_seconds = 2592000 }
  retention_policy { retention_period = 7776000 }
  encryption { default_kms_key_name = var.kms_key_name }
  labels = { purpose = "terraform-state", criticality = "tier0" }
}

# Grant only CI/CD state identities and audited break-glass recovery group.
resource "google_storage_bucket_iam_member" "state_ci" {
  bucket = google_storage_bucket.tfstate.name
  role   = "roles/storage.objectAdmin"
  member = "serviceAccount:terraform-deployer@${var.project_id}.iam.gserviceaccount.com"
}
```


19. Cross-Region Replication and Regional Failover

The enterprise uses paired recovery regions within each approved regional data plane. The pair is selected using legal residency, service availability, network connectivity, latency, quota/capacity, correlated hazard and cost. A region pair is not assumed to protect every service equally. Each workload documents which state is continuously replicated, periodically protected, rebuilt from source or intentionally unavailable in degraded mode.

Layer	Primary-to-recovery treatment
Identity and policy	Global or centrally managed identity with recovery groups, JIT privileges and policy as code; validate dependency availability.
Network	Predefined subnets, routes, Private Service Connect endpoints, firewall policy, DNS and hybrid connectivity; avoid overlapping emergency changes.
Data	Service-specific replication or protected recovery points with residency approval and measurable lag.
Compute/processing	Hot, warm, pilot-light or cold capacity by tier; quotas and images/templates prevalidated.
Orchestration	Recovery workflows and runbooks with primary fencing and explicit dependency order.
Consumer routing	Application routing, configuration or Cloud DNS; use read-only/degraded mode when data integrity is uncertain.
Operations	Regional dashboards, logs, alerts, incident channels, evidence repository and on-call coverage.

19.1 Residency and region pairing

Plane	Pairing principle	Constraint	Default pattern
AMER	Primary and paired recovery regions selected by latency, service availability, legal entity and risk assessment	No cross-border replication for locality-restricted products without legal approval	Tier 0/1 paired-region pattern; Tier 2 cold rebuild
EMEA	EU or country-approved region pair; use locality-restricted architecture where required	GDPR, sector regulation, records and works-council constraints	Prefer in-geography recovery; no global endpoint without residency validation
APAC	Country/market approved pair with service availability and network reachability	Data localization may restrict cross-country copies	Local pair where available; otherwise risk-accepted cold recovery or alternative service design
LATAM/Middle East	Map to approved regional plane or local deployment based on law and business criticality	Regional service gaps and cross-border transfer approval	Explicit exception and business-continuity workaround if paired services unavailable

20. DNS Failover

Cloud DNS routing policies and health checks can automate or assist failover for supported internal load balancers and external endpoints. Failover routing returns primary targets while healthy and backup targets when all primary targets are unhealthy. Geolocation routing can move traffic to the next closest healthy location when geofencing is disabled. DNS is only a routing layer; it must be coordinated with data activation, endpoint readiness and write fencing.

- Use TTL values that balance recovery speed with query load and client behavior; test real clients, resolvers, connection pools and caches.
- Use health checks that represent business readiness, not merely TCP reachability. A recovery endpoint is unhealthy until data, dependencies and security gates pass.
- Maintain a controlled manual override because automated health checks can misclassify partial failures.
- For APIs and AI endpoints, application-level routing or global load balancing may provide better session, authentication and observability control than DNS alone.
- Record every routing change in the incident evidence bundle and predefine rollback/failback steps.

20.1 Representative Terraform

```
# Purpose: Cloud DNS failover routing for supported regional internal load balancers.
# Validate endpoint type, global access, probing behavior, TTL and client caching before production use.
```



```

variable "project_id" { type = string }
variable "managed_zone" { type = string }
variable "dns_name" { type = string }
variable "primary_ip_address" { type = string }
variable "primary_port" { type = string }
variable "primary_protocol" { type = string, default = "tcp" }
variable "primary_network_url" { type = string }
variable "primary_region" { type = string }
variable "backup_region" { type = string }
variable "backup_ip_address" { type = string }

resource "google_dns_record_set" "service_failover" {
  project      = var.project_id
  managed_zone = var.managed_zone
  name         = var.dns_name
  type         = "A"
  ttl          = 30

  routing_policy {
    primary_backup {
      trickle_ratio = 0.01
    }

    primary {
      internal_load_balancers {
        load_balancer_type = "regionalL4ilb"
      }
    }
  }
}

```

21. Recovery Automation

Automation accelerates deterministic recovery tasks but does not remove accountability. Recovery workflows require an incident ID, approved change or emergency authorization, named target region, expected recovery point and validation callback. Automation identities are separate from normal deployment identities and receive only the permissions required for their steps.

Automation stage	Control
Precheck	Confirm declaration, approvals, region health, quotas, identity, keys, network, replication lag and trusted recovery point.
Fence	Stop primary schedulers/writes or establish epoch/lease ownership to prevent split brain.
Activate	Deploy/scale resources from pinned artifacts and restore/activate data in dependency order.
Validate	Run security, integrity, reconciliation, performance, freshness and dependency tests.
Release	Require explicit service-owner/business approval before traffic change.
Evidence	Persist commands, API outputs, logs, hashes, approvals, RTO/RPO and findings.
Rollback	Define safe stop and rollback for every automated action; do not improvise destructive reverse operations.

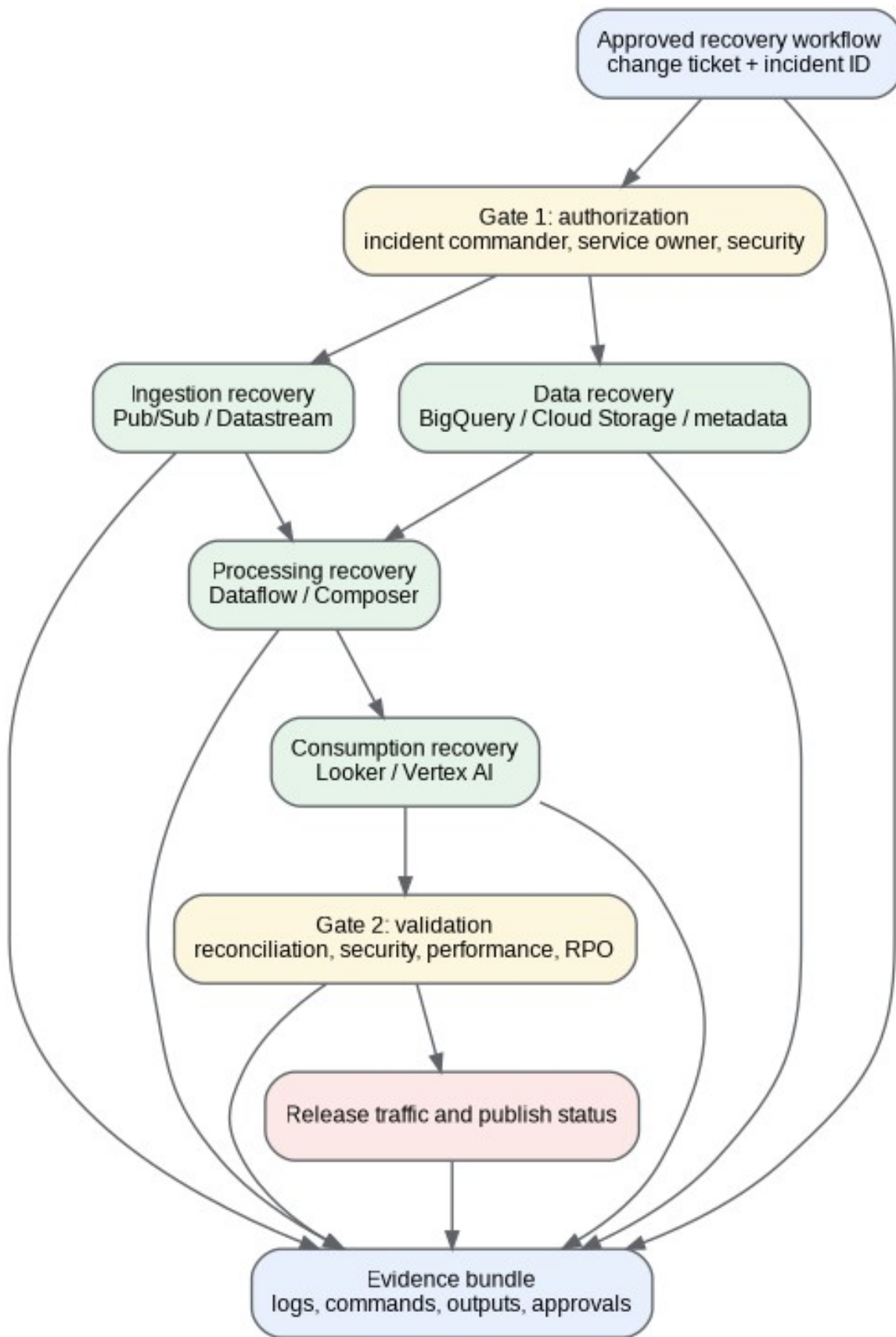


Figure 5 - Gated service recovery orchestration and evidence flow

21.1 Workflow starter

```
# Purpose: gated recovery orchestration skeleton for Google Cloud Workflows.
# Invoke only through an approved incident process and a least-privilege recovery service account.
main:
  params: [args]
  steps:
    - validate:
      switch:
        - condition: ${args.incident_id == null || args.change_id == null || args.approved != true}
          raise: "incident_id, change_id and approved=true are required"
    - record_start:
      call: sys.log
      args:
        severity: NOTICE
        text: ${"DR start incident=" + args.incident_id + " change=" + args.change_id}
    - prechecks:
      call: http.post
      args:
        url: ${args.precheck_url}
        auth: {type: OIDC}
        body: ${args}
      result: precheck
    - require_prechecks:
      switch:
        - condition: ${precheck.body.status != "PASS"}
          raise: '${"Prechecks failed: " + json.encode_to_string(precheck.body)}'
    - activate_data_plane:
      call: http.post
      args:
        url: ${args.activation_url}
        auth: {type: OIDC}
```

22. DR Runbooks

Every Tier 0/1 service has an executable runbook. The runbook is stored with source, rendered to an operator-friendly view, validated through exercises and linked from alerts and the service registry. It contains expected command outputs and stop conditions rather than only generic instructions.

Runbook field	Required content
Identification	service/workload ID, owner, tier, primary/recovery regions, version and last test
Trigger and authority	declaration criteria, incident commander, business decision maker and security escalation
Dependencies	identity, KMS, network, DNS, source, destinations, partners and human roles
Fencing	how to stop writes, schedulers, consumers and duplicate processing
Recovery point	how to select/approve it and expected RPO gap
Commands/automation	exact versioned scripts, parameters, identities, expected outputs and rollback
Validation	data control totals, quality, freshness, security, performance and business acceptance
Communication	audiences, templates, cadence and regulatory/legal approval
Failback	criteria, reverse synchronization, authoritative source and rollback
Evidence	repository, log links, approvals, actual RTO/RPO and findings

23. DR Testing and Exercises

Testing progresses from low-risk component restores to end-to-end regional and cyber exercises. Production failover is not the first time a team should discover missing quotas, unavailable keys, stale scripts or undocumented dependencies. Exercises are planned as changes, use realistic data and load where lawful, and produce auditable evidence.

Exercise	Scope	Cadence	Pass criterion
Runbook walkthrough	Tier 0/1	Quarterly	Steps current, contacts valid, dependencies confirmed
Backup restore sample	All protected services	Quarterly	Restore succeeds, integrity verified, evidence retained
Tabletop exercise	Tier 0/1 business service	Semiannual	Decision rights and communications work within target
Component failover	Tier 0/1	Semiannual	Component RTO/RPO and rollback proven
Regional failover exercise	Tier 0; selected Tier 1	Annual or after major change	End-to-end service restored and reconciled within target
Cyber recovery exercise	Regulated/Tier 0/1	Annual	Clean-room restore and security approval proven
Failback exercise	After failover design change or annual	Annual	No split brain, data reconciled, primary restored safely



Figure 6 - Continuous DR assurance and improvement cycle

23.1 Evidence and measurement

- Measure RTO from impairment to accepted minimum service and RPO from the last committed business record available after recovery.
- Record detection, declaration, fencing, restoration, validation and traffic-release timestamps separately to identify bottlenecks.
- Validate data completeness, duplicates, ordering, schema, policy, quality, financial/operational control totals and downstream behavior.
- Test operator access, break-glass, supplier escalation, customer/regulatory communication and executive decisions.
- All Critical and High findings block production approval or require time-bound residual-risk acceptance.

24. Failover and Failback

Failover makes the recovery environment authoritative. Failback is a separate controlled migration, not the automatic reverse of failover. The original primary can contain stale, partial or untrusted state. The incident commander, service owner and data owner decide the authoritative source before reverse synchronization.

Phase	Mandatory controls
Failover declaration	business impact, approved recovery point, write fencing, region capacity and communications
Recovery activation	dependency order, least-privilege identity, pinned artifacts and evidence
Service release	security, integrity, RPO, performance and business acceptance
Stabilization	heightened monitoring, backlog reduction, customer communication and change freeze
Failback planning	root cause resolved, primary rebuilt/clean, reverse data movement, downtime and rollback plan
Failback execution	fence recovery writes, final sync, validate primary, controlled routing and monitor
Closure	actual RTO/RPO, incident review, cost, findings, risk acceptance and architecture updates

25. Business Continuity

Business continuity defines how the enterprise sustains essential operations before full technology recovery. Each Tier 0/1 BIA identifies a minimum business service, maximum manual workload, alternate channel, staffing model, data

limitations and decision authority. Continuity procedures must explicitly prevent ungoverned spreadsheets, duplicate financial posting or privacy breaches during workarounds.

Continuity pattern	Use	Control
Read-only mode	Allow trusted historical lookup while writes are fenced	clear data timestamp and no implicit freshness guarantee
Delayed processing	Queue events/files until recovery	retention window, backlog capacity and customer expectation
Alternate channel	Use manual or secondary customer/partner process	identity, privacy, reconciliation and volume limits
Cached critical dataset	Serve pre-approved snapshot for operations	expiry, classification and prohibition on unsupported decisions
Regional workload transfer	Route work to unaffected operating center	staff authorization, time zone, residency and segregation
Manual priority processing	Handle highest-impact cases only	document selection criteria and later reconciliation

26. Crisis Communication

Crisis communication uses one verified incident record and separates internal operational updates, executive decisions, customer communication, partner notification and regulatory/legal reporting. Technical teams provide facts and uncertainty; authorized communications leaders approve external statements. Recovery estimates are expressed as ranges or next-update times until evidence supports a commitment.

Audience	Owner	Minimum content	Cadence
Technical responders	Incident commander	scope, actions, blockers, decisions, next checkpoint	15-30 minutes for active Tier 0
Executives/business owners	Crisis lead	business impact, RTO/RPO outlook, decisions and external exposure	30-60 minutes
Employees/operations	Business continuity	workaround, data limitations, safety/privacy controls and support channel	as process changes
Customers/partners	Authorized communications/business owner	verified impact, workaround, status and next update	contract and impact based
Regulators/legal/privacy	Legal/privacy/compliance	facts, affected records/subjects, containment and reporting obligations	statutory/contractual deadlines
Post-incident stakeholders	Service owner	root cause, actual RTO/RPO, data integrity, remediation and risk	approved final report

27. DR Governance and Operating Model

The BCDR Council owns policy, tier definitions, enterprise exercises, cross-service dependencies and residual-risk escalation. Platform and service teams own technical readiness and execution. Business/data-product owners own BIA, funding, validation and acceptance of service restoration. Security owns cyber-recovery controls and release approval after destructive incidents. Crisis Communications owns external messaging.

Forum	Cadence	Decisions/evidence
Service readiness review	Monthly for Tier 0/1	inventory, backup freshness, replication lag, runbook currency, findings and capacity
BCDR Council	Quarterly	tier changes, region pairs, exercise plan, residual risk and investment
Security/cyber recovery review	Quarterly and after threat changes	clean-room readiness, privileged access, trusted recovery points and evidence
Exercise review	After every exercise	actual RTO/RPO, integrity, people/process findings and remediation

Forum	Cadence	Decisions/evidence
Architecture review	After material change	service capability, dependency, region, data movement and automation design
Executive risk review	At least annual	enterprise exposure, unresolved High risks, funding and regulatory assurance

27.1 Governance RACI

Activity	BCDR	Platform/SRE	Business owner	Security	Data Eng	Analytics/AI
Maintain BIA and tier	A	C	R	C	C	I
Approve DR design	A	R	C	C	R	C
Operate backups and replication	C	A/R	R	R	C	C
Declare disaster	A/R	C	C	C	C	C
Execute technical recovery	C	A/R	R	R	R	R
Validate data and business service	C	R	A/R	C	C	C
Cyber recovery authorization	C	C	C	A/R	C	C
Crisis communication	A	I	R	C	I	I
Exercise and evidence	A	R	R	R	R	R
Accept residual risk	A	C	R	R	C	C

28. Structured Service Assessments

28.1 BigQuery

Assessment item	Enterprise response
Service name	BigQuery
Purpose and architecture role	Governed analytics data and critical data products
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Logical recovery through time travel/fail-safe and snapshots; regional recovery through Managed DR or dataset replication
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Managed DR and replication features, location pairs and compute readiness require validation Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Logical recovery through time travel/fail-safe and snapshots; regional recovery through Managed DR or dataset replication
Security, IAM and network	Dataset/table IAM, policy tags, row/column security, CMEK, VPC Service Controls where supported
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	Replication/standby compute, snapshots, data transfer and storage Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.

Assessment item	Enterprise response
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test RPO, routines/views/policies, reservations, consumer cutover and failback

28.2 Cloud Storage

Assessment item	Enterprise response
Service name	Cloud Storage
Purpose and architecture role	Landing, raw, replay, archive, artifacts and recovery evidence
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Dual/multi-region durability, turbo replication where eligible, soft delete, versioning and retention controls
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Replication is not independent backup; retention can conflict with deletion obligations Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Dual/multi-region durability, turbo replication where eligible, soft delete, versioning and retention controls
Security, IAM and network	Uniform bucket access, public access prevention, CMEK, VPC Service Controls, retention/holds
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	storage class, versions/soft-deleted bytes, replication and retrieval Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test object restore, generation integrity, policy, KMS and residency

28.3 Pub/Sub

Assessment item	Enterprise response
Service name	Pub/Sub
Purpose and architecture role	Durable event transport and replay
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Topic/subscription retention, snapshots and seek; recreate resources from IaC
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Replay window finite; duplicates and ordering effects possible Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Topic/subscription retention, snapshots and seek; recreate resources from IaC
Security, IAM and network	IAM, service accounts, CMEK where supported, VPC Service Controls and audit logs
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.

Assessment item	Enterprise response
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	retained bytes, throughput and recovery duplicate processing Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test seek, new subscription, fencing, idempotency, backlog and reconciliation

28.4 Dataflow

Assessment item	Enterprise response
Service name	Dataflow
Purpose and architecture role	Batch and streaming processing
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Recreate regional jobs from immutable templates; coordinate checkpoints, replay and sinks
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	No generic automatic cross-region job failover; stateful behavior is pipeline-specific Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Recreate regional jobs from immutable templates; coordinate checkpoints, replay and sinks
Security, IAM and network	Worker identity, private IPs, subnet, CMEK, VPC Service Controls and artifact trust
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	standby capacity, replay compute and cross-region transfer Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test template deployment, watermark, duplicates, sink integrity and rollback

28.5 Datastream

Assessment item	Enterprise response
Service name	Datastream
Purpose and architecture role	Database change data capture
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Resume after outage; stream recovery for source failover/log gaps; paired-region recreation where feasible
Benefits	Managed operation and integration with the inherited Google Cloud platform.

Assessment item	Enterprise response
Limitations and quotas	Regional outage pauses streams; duplicates may occur; source logs must remain available Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Resume after outage; stream recovery for source failover/log gaps; paired-region recreation where feasible
Security, IAM and network	Connection profiles, private connectivity, secrets/IAM, CMEK and audit logs
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	streaming volume, backfill and recovery transfer Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test source failover, log gap, backfill, duplicate removal and RPO

28.6 Managed Airflow / Cloud Composer

Assessment item	Enterprise response
Service name	Managed Airflow / Cloud Composer
Purpose and architecture role	Pipeline orchestration
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Scheduled environment snapshots plus Git/IaC-driven rebuild and paired environment
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Snapshots include sensitive database/fernet material and require a healthy environment to create Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Scheduled environment snapshots plus Git/IaC-driven rebuild and paired environment
Security, IAM and network	Private environment, Secret Manager, IAM, CMEK, VPC Service Controls and protected snapshot bucket
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	standby environment, snapshots, storage and network Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test snapshot save/load, DAG pause, dependencies, duplicate runs and schedules

28.7 Looker (Google Cloud core)

Assessment item	Enterprise response
Service name	Looker (Google Cloud core)
Purpose and architecture role	Governed BI and semantic consumption
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Same-instance backup restore; Git LookML and supported import/export for cross-instance recovery
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Backups may be instance-scoped; not all operational telemetry is included Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Same-instance backup restore; Git LookML and supported import/export for cross-instance recovery
Security, IAM and network	Instance IAM, OAuth/identity, database credentials, network and content permissions
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	standby instance, backup/export storage and capacity Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test instance/content recovery, connections, permissions, embeds and schedules

28.8 Vertex AI / Gemini

Assessment item	Enterprise response
Service name	Vertex AI / Gemini
Purpose and architecture role	ML/GenAI training, registry, endpoints, RAG and agents
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Versioned artifacts/configuration and paired-region deployment or degraded mode
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Location, model, endpoint and residency support vary; application state requires separate protection Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Versioned artifacts/configuration and paired-region deployment or degraded mode
Security, IAM and network	IAM, service accounts, private connectivity, CMEK where supported, safety and data controls
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	standby endpoint, model/artifact storage, quota and inference Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns

Assessment item	Enterprise response
	tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test deployability, prediction parity, quality/safety, RAG permissions and routing

28.9 Knowledge Catalog / Dataplex capabilities

Assessment item	Enterprise response
Service name	Knowledge Catalog / Dataplex capabilities
Purpose and architecture role	Metadata, glossary, quality, lineage and discovery
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Configuration as code, periodic inventories/exports and service rebuild where supported
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Do not assume all metadata has native cross-region backup Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Configuration as code, periodic inventories/exports and service rebuild where supported
Security, IAM and network	IAM, policy tags, VPC Service Controls, audit logs and least privilege
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	inventory/export storage and rebuild effort Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test glossary, owners, policy tags, quality rules, lineage and access controls

28.10 Cloud DNS

Assessment item	Enterprise response
Service name	Cloud DNS
Purpose and architecture role	Name resolution and failover routing
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Health-checked failover/geolocation policies plus manual override
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Client/resolver caching affects RTO; endpoint support is constrained; no data protection Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Health-checked failover/geolocation policies plus manual override
Security, IAM and network	DNS IAM, change logging, protected zones and controlled recovery identity
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location

Assessment item	Enterprise response
	support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	queries, health checks and operational overhead Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test health condition, TTL, real clients, manual failover and rollback

28.11 Cloud Storage for Terraform state

Assessment item	Enterprise response
Service name	Cloud Storage for Terraform state
Purpose and architecture role	Authoritative deployment state
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Versioning/soft delete, retention, restricted access and tested generation restore
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Restored state can diverge from real resources; locking/serial integrity required Quotas, limits and regional availability must be checked before implementation.
High availability and DR	Versioning/soft delete, retention, restricted access and tested generation restore
Security, IAM and network	Dedicated project, CMEK, no public access, JIT recovery and audit logs
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	versions, retained state and recovery copy Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Test exact generation restore, refresh-only plan and peer-approved apply

28.12 Cloud Workflows / Cloud Build

Assessment item	Enterprise response
Service name	Cloud Workflows / Cloud Build
Purpose and architecture role	Recovery orchestration and deployment execution
Business capabilities supported	Continuity of governed data, analytics, AI or platform control according to workload tier.
Technical capabilities	Versioned workflows/builds, recovery service accounts and evidence capture
Benefits	Managed operation and integration with the inherited Google Cloud platform.
Limitations and quotas	Automation can amplify incidents; regional dependencies and quotas must be considered Quotas, limits and regional availability must be checked before implementation.

Assessment item	Enterprise response
High availability and DR	Versioned workflows/builds, recovery service accounts and evidence capture
Security, IAM and network	least privilege, approvals, secrets, artifact provenance and audit logs
Encryption/CMEK and VPC Service Controls	Use inherited classification-based controls where the service and location support them; validate support.
Logging, monitoring and alerting	Route audit/operational logs and service metrics to Volume 11 observability; alert on backup freshness, lag, failures and recovery readiness.
Cost drivers and optimization	workflow/build execution and standby dependencies Follow Volume 10; resilience savings cannot create unapproved RTO/RPO regression.
Operational responsibilities	Central platform owns service baseline and automation; workload owner owns tier, data validation, runbook and exercise.
Alternatives and selection criteria	Choose native recovery features, source-based rebuild or third-party/alternate service only when evidence meets tier, residency, security and cost criteria.
Common pitfalls / anti-patterns	Treating HA as DR, replication as backup, untested scripts as recovery, or routing as data restoration.
Production readiness	Dry run, authorization gate, stop conditions, rollback and evidence completeness

29. Architecture Decision Records

The full ADRs are included as separate Markdown assets. The table summarizes the enterprise reference decisions; implementations can supersede them only through the documented architecture governance process.

ADR	Decision	Selected approach
ADR-1201	Adopt business-service-driven disaster recovery tiers	Classify workloads through BIA into Tier 0, Tier 1, Tier 2 and Tier 3, and bind architecture, testing and funding to approved RTO/RPO.
ADR-1202	Use paired regional data planes for Tier 0 and selected Tier 1 services	Establish residency-approved primary/recovery region pairs per AMER, EMEA and APAC data plane with workload-specific warm or hot patterns.
ADR-1203	Use BigQuery managed disaster recovery or tested cross-region replication for critical datasets	Use BigQuery Managed DR where edition, region pair and workload fit; otherwise use tested cross-region dataset replication plus explicit compute and consumer cutover procedures.
ADR-1204	Protect Cloud Storage with layered logical and geographic controls	Combine approved location strategy with soft delete, versioning where justified, retention/legal-hold controls and independent recovery evidence; use turbo replication selectively.
ADR-1205	Treat streaming recovery as replay plus idempotent processing	Retain messages, use snapshots/seek, fence consumers, design idempotent sinks and reconcile business keys after recovery.
ADR-1206	Recreate Dataflow and orchestration services from immutable artifacts and snapshots	Store deployable templates, DAGs, dependencies and configuration in immutable repositories; protect Composer snapshots and use tested regional rebuild procedures.
ADR-1207	Use clean-room cyber recovery for high-impact destructive incidents	Recover Tier 0/1 regulated services in isolated recovery projects with separate identities/keys, trusted restore points and security approval before re-entry.
ADR-1208	Use gated recovery automation rather than unrestricted one-click failover	Automate deterministic technical steps through least-privilege identities with incident/change identifiers, approval gates, fencing, validation and evidence capture.
ADR-1209	Use Cloud DNS failover only for compatible endpoints and tested client behavior	Use health-checked failover/geolocation policies where supported, with low but reasonable TTLs, application-level routing fallback and regular tests.
ADR-1210	Require periodic end-to-end exercises and evidence-based risk acceptance	Test restores quarterly, conduct Tier 0/1 table-tops semiannually, execute regional and cyber exercises at least annually, and track findings to closure or formal acceptance.

30. Risks and Mitigations

ID	Risk	Likelihood	Impact	Mitigation
DR-R01	Replication mistaken for backup	High	High	Independent logical recovery points, soft delete/versioning, cyber recovery copies and restore tests
DR-R02	Recovery region lacks quota or capacity	High	High	Quota reservations, capacity validation and pre-provisioned critical resources
DR-R03	Runbook succeeds technically but business data is inconsistent	High	High	Domain reconciliation rules, checkpoints, lineage and owner sign-off
DR-R04	CMEK or identity dependency unavailable during recovery	High	High	Key/identity dependency design, break-glass controls and exercises
DR-R05	Duplicate processing during replay/failover	Medium	High	Idempotency keys, deduplication, fencing and reconciliation
DR-R06	Malicious backups or contaminated recovery point	High	High	Immutable evidence, trusted-point selection, malware/integrity validation and clean room
DR-R07	DNS/client caching delays failover	Medium	Medium	Appropriate TTL, health checks, client behavior tests and manual routing option
DR-R08	Managed service feature or location mismatch	Medium	High	Current capability validation, service-specific tests and documented fallback
DR-R09	Recovery automation has excessive privilege	High	High	Separate recovery identities, JIT approval, dry run and audit logging
DR-R10	Fallback causes split-brain or data loss	High	High	Write fencing, authoritative source decision, reverse sync and controlled validation

31. Implementation Roadmap

Phase	Indicative duration	Key outcomes	Exit gate
0 - Mobilize and inventory	4-6 weeks	Service inventory, BIA template, owners, tiers, dependencies, current backup and region assessment	Tier 0/1 scope approved
1 - Recovery foundation	6-10 weeks	Region pairs, recovery projects, identities, keys, networks, source/artifact/state protection and evidence repository	Foundation restore test passes
2 - Critical data protection	8-12 weeks	BigQuery and Cloud Storage patterns, Pub/Sub retention, CDC recovery and metadata protection	Tier 0/1 data RPO demonstrated
3 - Runtime and consumer recovery	8-12 weeks	Dataflow, Composer, Looker, AI, DNS/application routing and recovery automation	Service recovery tests pass
4 - Cyber recovery and continuity	6-10 weeks	Clean-room design, crisis communications, manual/degraded processes and legal/privacy integration	Cyber tabletop and restore pass
5 - Exercise and scale	Ongoing	Regional exercises, findings remediation, domain onboarding, assurance dashboard and annual risk review	Executive BCDR acceptance

31.1 Prioritized first 90 days

- Approve tier policy, BIA template, RTO/RPO measurement definitions and decision authority.
- Inventory Tier 0/1 services and map identity, KMS, network, source, data, pipelines, consumers and external dependencies.
- Choose initial AMER, EMEA and APAC recovery pairs and validate service availability, quotas and residency.
- Protect Terraform state, source, artifacts, Composer snapshots and critical Cloud Storage/BigQuery data.
- Build one end-to-end pilot covering Pub/Sub/Dataflow/BigQuery/Looker or an equivalent critical path.
- Conduct a tabletop and component restore; publish measured gaps and funded remediation plan.

32. Validation Criteria and Acceptance Tests

ID	Acceptance criterion	Validation method
VAL-01	All Tier 0/1 services have approved BIA, owner, tier, dependency map, region pair and runbook	Service registry query and governance approval
VAL-02	Backups/snapshots/replication meet freshness and retention policy	Automated control report and sample restore
VAL-03	Tier 0 end-to-end recovery completes under 60 minutes with less than 5 minutes data loss	Exercise evidence and business validation
VAL-04	Tier 1 end-to-end recovery completes under 4 hours with less than 15 minutes data loss	Exercise evidence and business validation
VAL-05	Data replay does not create unaccepted duplicates or financial/operational imbalance	SQL reconciliation and owner sign-off
VAL-06	Recovery works with primary-region, normal deployer and compromised credentials unavailable	Controlled exercise using recovery identities
VAL-07	CMEK, service perimeter, private connectivity and least privilege work in recovery region	Security test and audit logs
VAL-08	DNS/application cutover is observed by representative clients within target	Synthetic and real-client test
VAL-09	Failback completes without split brain and with authoritative data reconciliation	Failback exercise report
VAL-10	Crisis communication, legal/privacy escalation and continuity workaround meet agreed cadence	Tabletop evidence
VAL-11	All High/Critical findings are closed or time-bound risk accepted	Risk register and approval
VAL-12	Recovery evidence schema is populated and stored immutably for every exercise	Evidence repository audit

33. Production Readiness Checklist

- ☐ PR-01 Business owner and service owner are named and reachable through on-call/escalation paths.
- ☐ PR-02 BIA, tier, RTO, RPO, minimum operating service and continuity workaround are approved.
- ☐ PR-03 Primary and recovery regions comply with residency and have validated service support, quotas and capacity.
- ☐ PR-04 Identity, KMS, network, DNS, hybrid connectivity, service perimeter and source dependencies are included.
- ☐ PR-05 Backups, snapshots, time travel, soft delete/versioning and replication are configured and monitored as designed.
- ☐ PR-06 Terraform state, source, artifacts, DAGs, LookML, prompts/models and configuration can be recovered without the primary region.
- ☐ PR-07 Runbook includes fencing, trusted recovery point, exact commands, stop conditions, validation, communication and failback.
- ☐ PR-08 Recovery automation uses separate least-privilege identities, approvals, incident IDs, logging and rollback.
- ☐ PR-09 Data validation covers freshness, completeness, duplicates, ordering, schema, access policy and business control totals.

- ☐ PR-10 Degraded mode and manual processes have privacy, security, reconciliation, staffing and capacity controls.
- ☐ PR-11 Component restore and end-to-end exercise meet target or have an approved time-bound exception.
- ☐ PR-12 Actual RTO/RPO measurement and evidence retention are automated or unambiguous.
- ☐ PR-13 Cyber recovery design identifies trusted recovery points, clean-room projects, keys, identities and release authority.
- ☐ PR-14 Failback and post-incident review are tested and funded.
- ☐ PR-15 FinOps assessment covers replication, storage, standby, data transfer, exercise and recovery-operation costs.

34. Dependencies, Decisions and Next Actions

34.1 Cross-volume dependencies

Volume	Dependency
01 Executive Architecture	Business outcomes, service-tier policy, global region strategy and decision authority.
02 Landing Zone	Organization, projects, Shared VPC, identity, KMS, VPC Service Controls, logging and regional foundations.
03 Data Platform	Data layers, BigQuery/Cloud Storage architecture, metadata, security and data-product SLOs.
04 Data Engineering	Idempotency, replay, backfill, checkpoints, templates and operational standards.
05 Data Mesh	Domain ownership, product contracts, SLOs and business validation.
06 Analytics	Looker semantic layer, dashboards, access and analytics SLOs.
07 AI/GenAI	Model, prompt, RAG, agent, evaluation and AI governance artifacts.
08 Security	Zero Trust, incident response, keys, privileged access, cyber controls and evidence.
09 DevSecOps	Source, artifacts, CI/CD, promotion, rollback and supply-chain assurance.
10 FinOps	Cost model for replication, backups, standby capacity, exercises and recovery operations.
11 SRE/Operations	SLIs/SLOs, alerting, incident command, on-call, runbooks and observability.
13 Terraform	Reusable modules, remote state, deployment and drift recovery.

34.2 Executive decisions required

- Approve the BIA-driven Tier 0/1/2/3 model and residual-risk authority.
- Approve recovery region pairs and locality-restricted exceptions for each data plane.
- Fund Tier 0/1 replication, standby capacity, exercises and cyber-recovery foundation.
- Designate disaster declaration, traffic release, data-integrity and external-communication authorities.
- Require annual end-to-end regional and cyber exercises for critical services.

34.3 Next actions

- Run a 30-day Tier 0/1 inventory and BIA campaign.
- Pilot BigQuery, Cloud Storage, Pub/Sub, Dataflow and Looker recovery for one high-value data product.
- Create the recovery project/identity/key/network foundation and protect Terraform state, source and artifacts.
- Implement the evidence schema and executive DR readiness dashboard.
- Schedule the first cross-functional tabletop within 60 days and technical failover exercise within 120 days.

35. References and Companion Assets

Source	Use in this volume
Google Cloud Architecture Center	Disaster recovery planning guide and disaster recovery scenarios for data/applications.
BigQuery documentation	Managed disaster recovery, cross-region dataset replication, reliability,

Source	Use in this volume
	time travel/fail-safe, table snapshots and dataset management.
Cloud Storage documentation	Data protection, backup and recovery options; availability/durability; turbo replication; soft delete; consistency and storage classes.
Pub/Sub documentation	Reliability introduction, replay and purge with seek, snapshots, retention and Dataflow integration.
Dataflow documentation	Streaming with Pub/Sub, templates/updates, regional endpoints, snapshot and CMEK considerations.
Datastream documentation	Behavior overview, event delivery and stream recovery.
Managed Airflow / Cloud Composer documentation	Save/load environment snapshots, scheduled snapshots and disaster recovery with snapshots.
Looker documentation	Looker (Google Cloud core) backup/restore and supported import/export or customer-hosted restore guidance.
Cloud DNS documentation	Routing policies, health checks and failover configuration.
Vertex AI / Gemini documentation	Locations, model registry, deployment, generative AI locations and data residency; validate current naming and support.
Enterprise Blueprint Series	Volumes 01-10 and master prompt; inherited architecture, controls and terminology.

Companion package assets include six diagrams in DOT, Mermaid, PlantUML, SVG, PNG and draw.io formats; implementation starters; eight CSV matrices; ten detailed ADRs; a manifest, checksum file and validation report. Examples are reference starters and require project-specific testing, security review and current Google Cloud capability validation before production use.